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By Stanford ML Group

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Tech Overview

BRL Architecture

 **Could you recommend journals that would be a good fit for robotics-related research?**

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Summary

This paper presents a knowledge-based decision-making framework that enables a robot to decide when to ask for help and what to ask by maintaining a belief over symbolic world hypotheses and querying humans at the state (predicate) level when ambiguity remains. The approach formulates execution as a POMDP coupled with a STRIPS-style model, uses a POMCP planner constrained to applicable symbolic actions, and selects queries to minimize expected entropy over a particle belief defined on the reachable “frontier” of successor states. The authors integrate sensing, planning, feedback management, and a user interface, and evaluate the system in waste sorting and tomato harvesting via system experiments and a small Wizard-of-Oz user study, showing improved task success versus no/naïve querying and reduced interactions versus dense supervision.

Strengths

Technical novelty and innovation

- Integrates symbolic STRIPS models with POMDP-style online planning (POMCP) and state-level human queries, tying “when” and “what” to ask directly to the planning belief.
- Introduces a frontier-based symbolic belief approximation that focuses inference on reachable successor worlds, reducing combinatorial blow-up typical of global symbolic beliefs.
- Proposes an expected-entropy query selection at the predicate level, producing interpretable, user-verifiable questions grounded in the knowledge base.
- Provides a clean execution loop that commits high-confidence hypotheses to the KB and requests targeted feedback otherwise, closing the loop between belief, planning, and interaction.

Experimental rigor and validation

- Systematic threshold ablation explores the trade-off between success and interaction cost across domains, offering practical guidance for setting the confidence threshold.
- Baseline comparisons include no queries, random queries, dense (“All”) supervision, and an action-level KnowNo-style baseline; results are presented with success, queries, query rate, and planning length.
- Scalability analysis increases object count and reports not only performance but also computational indicators (frontier size, nodes expanded, episode time).
- A Wizard-of-Oz user study (n=12) measures task success, operation time, workload (NASA TLX), and situation awareness (SAGAT), complementing system-level results.

Clarity of presentation

- The system architecture and data flow are clearly explained, with consistent notation and intuitive figures linking KB, planner, sensing, and feedback.
- Preliminaries concisely recap STRIPS, POMDPs, and POMCP and explain how symbolic constraints restrict search/action feasibility.

- The uncertainty/confidence formulation and expected-entropy-based query selection are well-motivated and described with precise equations.

Significance of contributions

- Addresses a central HRI problem—when and what to ask—at execution time, demonstrating fewer queries than dense supervision while maintaining high success in two distinct domains.
- Presents a practical, interpretable alternative to action-level queries that shifts the cognitive burden from action selection to fact verification, likely improving operator workload and calibration.

Weaknesses

Technical limitations or concerns

- The belief is defined on a per-action “frontier” rather than a persistent global belief over all latent symbolic facts, which may be myopic and can fail if the correct world falls outside the enumerated frontier; the corollary assumes complete frontier coverage and perfect feedback.
- The framework commits to a single knowledge state once confidence exceeds a threshold but lacks mechanisms for backtracking or belief recovery if observations/models are misspecified or human feedback is noisy.
- Query selection optimizes expected entropy (information gain) rather than expected value-of-information relative to task reward; this can favor information that is not utility-critical.
- Human error is not modeled; no analysis of robustness to noisy or delayed feedback, despite known variability in operator inputs.

Experimental gaps or methodological issues

- The KnowNo action-query baseline differs semantically from the proposed state-level feedback; the comparison is informative but not apples-to-apples, and implementation details (e.g., prompting, access to internal state) may affect fairness.
- The user study is small (n=12), Wizard-of-Oz, and exploratory (no multiple-comparison correction), limiting strength of user-facing claims;

also, tomato KnowNo achieved 100% success in WoZ versus low system-level success, suggesting condition or setup mismatch.

- No explicit modeling of query costs in the reward; reliance on a confidence threshold τ may conflate interaction cost with belief sharpness.
- Sensitivity to observation/transition probability misspecification is acknowledged but not experimentally studied; these parameters strongly affect confidence and query frequency.

Clarity or presentation issues

- Minor inconsistencies in figure captions/axis labels (e.g., plots labeled “Query Probability per Step” under different panels) may confuse readers; some formatting artifacts (from PDF extraction) persist.
- Some implementation specifics for observation likelihoods are deferred to appendices, which are crucial to reproduce the results.

Missing related work or comparisons

- Limited discussion of contingent/conditional planning and sensing within symbolic planning (e.g., conditional/knowledge-level planning in PDDL, contingent planners, mixed observability planning frameworks) that also address uncertainty and information-gathering actions.
- Prior work on value-of-information and VOC/VPI-based query selection in planning under uncertainty is not compared, despite conceptual proximity to the proposed entropy-based strategy.

Detailed Comments

Technical soundness evaluation

- The integration of STRIPS constraints into POMCP by restricting action applicability is sound and improves search efficiency.
- The frontier-based belief approximation is a practical design that exploits symbolic structure; however, it assumes the frontier covers all materially distinct next-world hypotheses. Violations (e.g., under-modeled outcomes) can lead to erroneous commitments without recovery.

- The confidence metric (entropy normalized by $\log_2$ of frontier size) is reasonable for thresholding but may not correlate with downstream value; a VOI-based trigger could better reflect utility.
- The query selection criterion (expected entropy after querying) is standard and appropriate but myopic; tasks with long dependencies (e.g., tomato harvesting) may benefit from non-myopic or reward-weighted VOI.

Experimental evaluation assessment

- The threshold ablation is thorough and illuminates domain differences, showing that over-querying yields diminishing returns and that long-horizon tasks require higher confidence.
- Baseline comparison convincingly demonstrates that naïve or random querying underperforms targeted queries, and that dense supervision inflates interaction cost; however, including a VOI-based state-query baseline or a contingent planner with sensing actions would strengthen the case.
- Scalability tests are valuable and candid: they report increased runtime/frontier sizes and reduced success in more complex tomato scenes, clarifying computational limitations.
- The WoZ user study provides early evidence of usability and workload advantages, but results should be interpreted cautiously due to sample size, scenario scripts, and the lack of end-to-end autonomy during trials.

Comparison with related work (using the summaries provided)

- Relative to KnowNo-style methods that ask action-level questions, this work's state-level queries reduce the operator's SA burden and empirically improve success in system tests, aligning with the claim that fact verification is easier than action selection under uncertainty.
- Compared to prior symbolic uncertainty planning and ASP/POMDP hybrids, the proposed frontier belief and online POMCP integration offer a pragmatic alternative that emphasizes interpretable facts and human-in-the-loop disambiguation rather than heavy offline model construction.
- LLM-based planning methods provide flexible task interpretations but struggle with executability and uncertainty; here, symbolic transition models ensure feasibility, and uncertainty is handled explicitly with a belief and observation model.

- The paper could better position itself with respect to contingent planning and VOI-based active perception, which aim to select informative queries/sensing actions grounded in task reward.

Discussion of broader impact and significance

- The framework promotes transparent robot behavior by exposing uncertainty as symbolic predicates, facilitating reliable human correction and potentially improving trust/calibration.
- In real deployments, modeling human error and query cost—plus adding backtracking—will be essential to prevent lock-in to incorrect knowledge states and to bound cognitive load.
- The architecture is general and could transfer beyond the two domains provided appropriate predicates, action models, and observation likelihoods are defined; however, engineering effort for model specification and calibration may be substantial.

Questions for Authors

1. How robust is the framework to erroneous human responses or misspecified observation/transition probabilities? Can you report results with injected label noise or perturbed model parameters to quantify degradation and recovery?
2. Does the system support backtracking or belief revision after committing to a knowledge state if later evidence contradicts the commitment? If not, how might you extend the approach to allow reversible commitments or maintain a low-weight alternative hypothesis set?
3. Why was expected entropy chosen over a reward-weighted value-of-information criterion? Have you evaluated a VOI trigger and/or query selection that accounts for task utility rather than pure information gain?
4. How is the frontier constructed for complex, multi-object operators where combinatorial effects (e.g., occlusions, mispicks) proliferate? Are there pruning heuristics or learned surrogates to keep frontier sizes tractable?
5. Could you clarify the KnowNo baseline setup: what information (if any) about the current belief/state was exposed to the user, and how were prompts/actions standardized across trials to ensure fairness? How do you

reconcile the large discrepancy between system-level and WoZ KnowNo performance?

6. What is the runtime budget per decision step on the real robot, and which components dominate latency (POMCP simulations, SP likelihoods, UI/feedback)? Any avenues to accelerate the planner for larger scenes?
7. How portable is the observation model specification across domains? Can you provide guidance or tools to help practitioners elicit/estimate $O(o|s,a)$ from data to reduce manual tuning?

Overall Assessment

This paper tackles a timely and important HRI problem—deciding when and what to ask under uncertainty—by tightly coupling symbolic planning with a belief-driven, state-level querying mechanism. The architecture is coherent and well-implemented, and the empirical results, especially the threshold ablations and scalability study, provide meaningful evidence that targeted state-level questions can preserve high success with substantially fewer interactions than dense supervision. The main limitations lie in the myopic frontier belief, the lack of robustness analyses (human error, model misspecification), and the absence of explicit VOI-style baselines or contingent planning comparisons. The user study is informative but small and exploratory. Overall, the work is a solid systems contribution with moderate novelty and clear practical value; addressing robustness and adding reward-aware query baselines would strengthen it further. I recommend publication at Could you recommend journals that would be a good fit for robotics-related research? after minor-to-moderate revisions focused on robustness experiments, baseline breadth, and clarifications of the baseline setups and limitations.

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